Does Graph Smoothing Reintroduce Bias into Fair Node Embeddings?

ICT 5143: AFGML Capstone | Problem 4

[Student 1, roll number] [Student 2, roll number]

[Evaluation date]

Research question and decision context

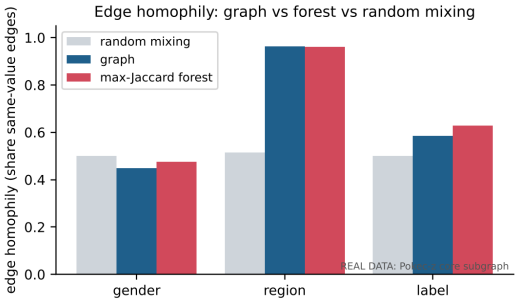
- Offline audit of node classification on Pokec-z (FairGNN release); not a recruitment system.
- $Y = 1$: benchmark's positive working-field class (raw code ≥ 1, meaning undocumented). $D = 1$: pipeline predicts that class.
- **Question:** when does probabilistic graph smoothing undo the fairness gains of debiased embeddings, and can constrained edge potentials reduce this effect?
- **Hypothesis:** with demographic homophily + base-rate gap, uniform smoothing widens the gender TPR gap (ΔEOpp); falsified if paired block-bootstrap effects are ≤ 0 / not significant after Holm.

Closest work and proposed contribution

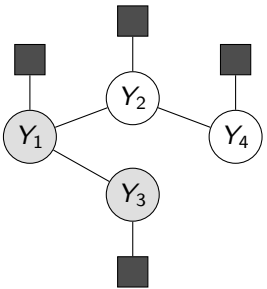
- **Established:** FairWalk debiases walks (Rahman et al. 2019, reproduced); FairDrop/CrossWalk reweight homophilous edges during representation learning; Correct&Smooth smooths a base predictor, no fairness analysis; FairPSL: fairness in relational MAP inference.
- **Open (within 12 verified papers):** fairness change *between* a fixed debiased embedding and a downstream exact CMN, under controlled homophily.
- **Contribution:** exact-inference decomposition + homophily-constrained attenuation (HCA) $\lambda_{ij} = \lambda(1 - \eta\,\mathbf{1}[A_i{=}A_j])$.
- **Evidence needed:** HCA must beat a *global scalar* with the same tuning budget (H4); otherwise it is just "smooth less".

Data and evaluation population

- Fixed core: 2,000 labelled nodes, 5,351 edges (BFS, seed 20261105, label-value blind).
- $P(Y{=}1)$: gender 0 **0.40**, gender 1 **0.56**.
- Edge homophily: gender 0.449 (random 0.500), region 0.963, label 0.586.
- Transductive 50/25/25 splits $\times$ 5 seeds; test cells $\geq$33 per seed.
- Gender: recorded code only, never inferred.



Graphical model and inference



grey = observed training label; squares = unary $\psi_i(Y_i; H_i)$; edges = ψ_{ij} on forest E_F

$$P(\mathbf{Y} \mid \mathbf{H}, E_F) = \tfrac{1}{Z} \prod_i \psi_i(Y_i) \prod_{(i,j) \in E_F} e^{\lambda_{ij} \mathbf{1}[Y_i = Y_j]}$$

- Query: $q_i = P(Y_i{=}1 \mid \mathbf{y}_{\text{train}})$, exact log-space sum-product on max-Jaccard forest (37.4% of edges).
- Markov blanket = forest neighbours; each positive neighbour adds λ_{ij} to log-odds.
- Validation: enumeration = sum-product = VE = clique tree, max error 1.6e-15 on 200 fixtures (tol. 10^{-8}).

Fairness criterion and proposed method

- **Primary:** gender $\Delta\text{EOpp} = \max_g \mathrm{TPR}_g - \min_g \mathrm{TPR}_g$: base rates of the recorded label differ, the risk is unequal errors among true positives. Diagnostics: ΔDP, ΔEOdds, R_{sel}, intersectional, calibration.
- **Utility:** macro-F1; acceptable loss 0.02 (validation).
- **HCA:** λ fit on training folds (held-out log-likelihood); $\eta = \text{argmin val } \Delta\text{EOpp}$ s.t. val F1 $\geq$ uniform -0.02.
- **Protected attribute at prediction time:** HCA yes (edge weights); FairWalk only for walks; group thresholds yes.

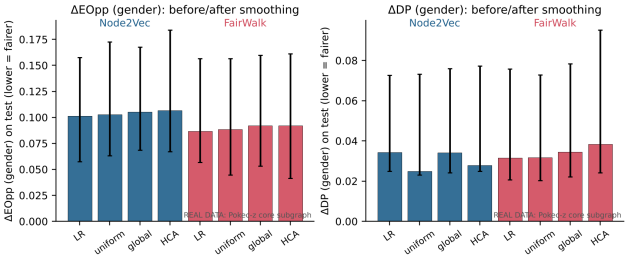
Experimental protocol

- Baselines: feature LR, SGC, Node2Vec+LR, FairWalk+LR, group thresholds, uniform CMN; ablations: no pairwise, global scalar, Node2Vec for FairWalk.
- Same graph, splits, forest, threshold rule and 11-point grids for every method.
- 5 seeds (training variability) **separately from** 1,000 Louvain-block bootstrap replicates (15 blocks; evaluation uncertainty) and 10 synthetic realisations/cell.
- Confirmatory H1–H4 (Holm), declared before test evaluation; all else exploratory.
- Exp. 4: simulated homophily $\times$ degree imbalance; Exp. 5 + robustness: dropout, corruption, rewiring, 4 forests, 3 dims, 4 label budgets.

Main result and uncertainty

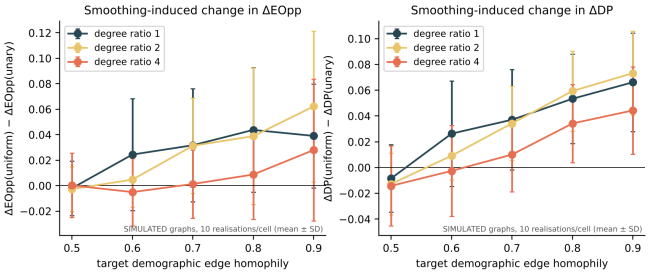
Method	F1	ΔEOpp	Brier
Node2Vec+LR	0.667	0.101	0.214
FairWalk+LR	0.657	0.087	0.215
FW+uniform	0.669	0.089	0.211
FW+global	0.660	0.092	0.214
FW+HCA	0.660	0.092	0.212
N2V+thresholds	0.667	0.013	0.214

H1 uniform−LR (FW): 0.002 [-0.038, 0.022], Holm p=1.000
H4 HCA−global: 0.000 [-0.037, 0.023], Holm p=1.000



Answer: on this subgraph smoothing neither undid nor created a gender gap; HCA ≈ global scalar.

When does smoothing reintroduce bias? (SIMULATED)



- Unary fair by construction; uniform $\lambda = 1$.
- ΔDP change: 0.066 at homophily 0.9 vs -0.009 at 0.5 (SD over 10 realisations).
- No base-rate gap: only 0.014 at 0.9. HCA−global: -0.006 to 0.020.
- Pokec-z gender sits at the no-homophily end (0.449), consistent with the real-data null.
- Simulated mechanism $\neq$ fact about Pokec.

Ablation, robustness and failure case

- **Ablation A:** no pairwise factors reproduces unary predictions exactly ($\Delta = 0$).
- **Ablation B:** HCA vs global scalar 0.000 [-0.037, 0.023] — no evidence group-aware attenuation helps.
- **Ablation C:** Node2Vec HCA−uniform 0.004 [-0.013, 0.020].
- **Robustness** (15 conditions): smoothing effect on ΔEOpp −0.027 to 0.019 (FairWalk), no consistent sign; F1 gain needs the Jaccard forest.
- **Failure:** validation-selected η unstable across seeds (0.8, 0, 0.3, 0.5, 1); weak fitted coupling $\lambda = 0.25$.
- Decision thresholds: ΔEOpp 0.013 but ΔDP 0.098 — criteria conflict.

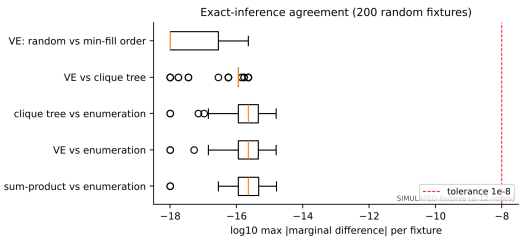
Conclusion and reproducibility

- **Conclusion:** smoothing reintroduces bias only when neighbourhoods carry group base rates; not the case for gender on this Pokec-z subgraph. HCA gives no gain over weaker smoothing.
- **Limits/ethics:** one subgraph; region leaks from embeddings (AUROC 0.991); HCA is differential edge treatment; label has no employment use.
- **Reproduce:** `scripts/run_experiments.py`, `evaluate.py`, `make_results.py`; commit [hash].
- Student 1: [contribution and verification]. Student 2: [contribution and verification].

Backup: mandatory evidence

Requirement	Artifact
Exp. 1 Node2Vec reproduction	`results/metrics/main/`, Fig. 6, 8
Exp. 2 uniform smoothing, both embeddings	H1, H2, Fig. 7
Exp. 3 attenuation vs embedding/decision mitigation	H3, H4, E6, E7, Fig. 9
Exp. 4 synthetic homophily/degree	Fig. 11, 12
Exp. 5 missing/corrupted edges, forest loss	Fig. 13, 14
Ablations A–C	`table_ablations_and_comparisons.csv`
Robustness (5 tests)	`table_robustness*.csv`
Full matrix	`docs/requirements_matrix.md`

Backup: inference and statistical validation



- Tests: factor scopes, indexing, evidence, normalisation, isolated nodes, components, zero mass, joint CI $\{0,1\} \perp \{3,4,5\} \mid 2$, Markov blanket vs enumeration.
- Bootstrap: Louvain blocks, identical draws across methods, inter-block edge share 0.264.
- Holm over H1–H4; effect sizes = paired differences + per-seed range.

Backup: demonstration and individual defence

- Live: `python scripts/smoke_test.py` (30 s, simulated) and `python scripts/verify_inference.py` (10 s).
- Real: `run_experiments.py -seeds 11 -n-jobs 1` (4 min).
- Fallback: `demo/saved_trace.txt`.
- Each student: trace one message $m_{u \to v}$, justify EOpp over DP, critique the HCA claim.